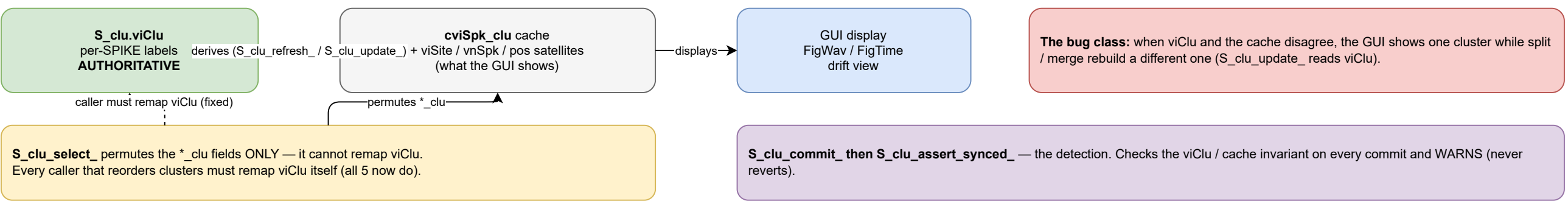


IronClust cluster-identity bugs — where they were, how they were fixed, what the code does now

Green = fixed & committed (87cd4f1, d954926) · Purple = detection added · Orange = still open (plan P3b)

How the code works now (the invariant): `viClu(cviSpk_clu{i}) == i`



The bugs — location · defect · fix · current behavior

CID-01 · reorder_clu_by_coords_ <i>irc.m 10380</i> Bug [O] renumbered the cache, not viClu, then saved. Fix remap viClu before S_clu_select_. Now [O] moves cache + viClu together.	CID-07 · delete_clu_ <i>irc.m 9140</i> Bug remapped viClu even when the cache remap silently failed (7,172 deleted spikes found in live cache). Fix snapshot + content-check + rollback. Now a failed remap aborts cleanly.	CID-03 · split_clu_ <i>irc.m 10690</i> Bug max(viClu)+1 clobbered a live index; the split silently reverted. Fix max(nClu, max(viClu)) + 1. Now the split always commits.	CID-04 · get_clu_spk_confirmed_ <i>irc.m 9873</i> Bug on total disagreement it returned the stale cache (why 45b5333 did not fix it). Fix prefer viClu (authoritative). Now splits use the right spikes even when desynced.	CID-02 · show_drift_view [S] <i>show_drift_view.m</i> Bug polygon drawn on gca, data read from the tab axes. Fix impoly_(hAx) + explicit parent. Now the cut matches what is drawn.
CID-08 · merge_clu_ <i>irc.m 9344</i> Bug a half-applied merge was logged as complete. Fix snapshot + whole-merge rollback. Now merge is all-or-nothing.	CID-09 · post_merge_wav_ <i>irc.m 4287</i> Bug stripped mrWavCor and left nClu_merge unassigned, crashing the two-output caller. Fix init nClu_merge=0; return before rmfield. Now fSave_spkwav=0 is a true no-op.	CID-10 · parpool sizing <i>irc.m 2572</i> Bug a stale undersized pool was reused, running the loop narrow. Fix resize in either direction (~= nWorkers). Now the pool matches the requested width.	CID-06 · S_clu_assert_synced_ <i>irc.m 19787 (detection)</i> Gap no content check existed (S_clu_valid_ checks lengths only). Fix new O(nSpk) check at S_clu_commit_, warns. Now a desync is announced, not hidden.	CID-11 · struct_select_safe_ <i>irc.m 19589 (ENABLER — OPEN)</i> Enabler skips a field silently; the reconcile then falsifies the length. Status NOT fixed (plan P3b). Mitigated delete_clu_'s content guard catches it locally.